📈 Curved Graphs — IGCSE Mathematics 0580

Hi there! 👏 A curve is just a **collection of points** joined together smoothly. Plot some points, join them up, and discover the secrets of each curve. Take your time — you've got this!

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⭐ **Score: 0 / 63** — 1 point for every correct answer (table values and practice questions)!

Quadratic Graphs: $y = ax^2 + bx + c$

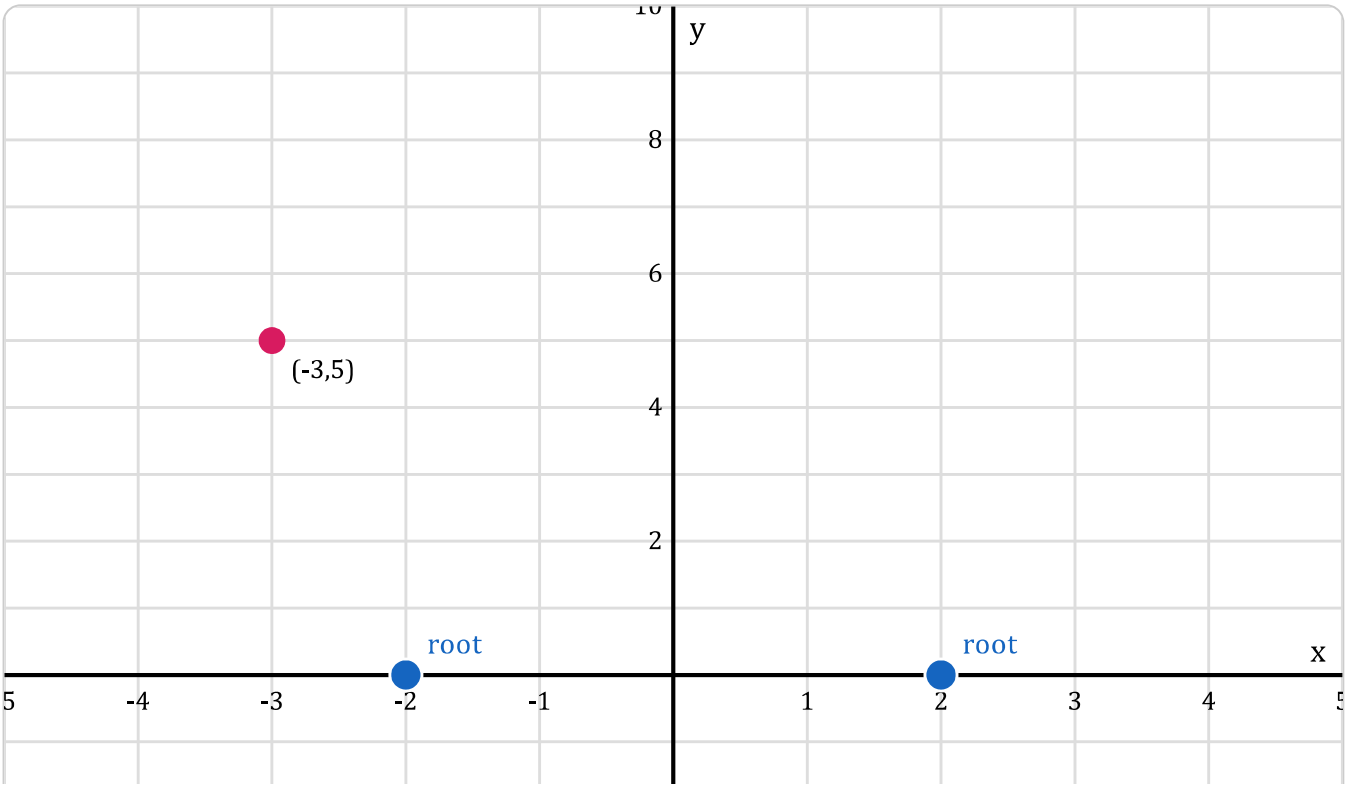
A quadratic makes a **U-shape** (a "parabola"). If a is positive it opens up ∪; if a is negative it opens down ∩.

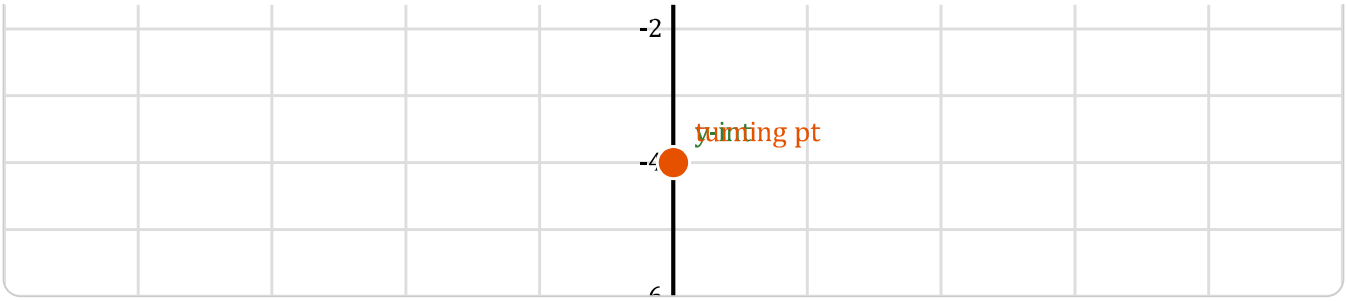
Choose an equation: $y = x^2 - 4$

📝 **Complete the table:** work out each **y** by putting the x-value into the equation. The first one is done for you. Each correct answer plots a point on the graph!

x	-3	-2	-1	0	1	2	3
y	5						

☐ **Connect Points** (unlocks when your table is complete)





Show features: ☑ Roots ☑ y-intercept ☑ Turning points ☑ Asymptotes

> **Roots**: where the curve crosses the x-axis (y = 0).

> **y-intercept**: where it crosses the y-axis (x = 0).

> **Turning point**: the lowest (or highest) point where the curve changes direction. Left of it the gradient is negative (going down); right of it, positive (going up) — or the reverse if it opens down.

> Quadratics have no asymptotes.

My Graph Analysis & Observations

Tap a sentence starter, then finish it:

Write what you notice…

📝 Practice Checkpoint

1. For $y = x^2 - 4$, find y when x = 3.

Answer:

Show your steps:

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2. What is the y-intercept of $y = x^2 - 4$?

Answer:

Show your steps:

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3. What is the smaller root of $y = x^2 - 4$?

Answer:

Show your steps:

Cubic Graphs: $y = ax^3 + bx^2 + cx + d$

A cubic makes an **S-shape** wiggle. It can have up to 3 roots and up to 2 turning points (one peak, one trough).

Choose an equation: $y = x^3 - 4x$

Complete the table: work out each **y** by putting the x-value into the equation. The first one is done for you. Each correct answer plots a point on the graph!

x	-3	-2	-1	0	1	2	3
y	-15						

Connect Points (unlocks when your table is complete)

